Full Title: AI4Nature: Computer Vision and AI for Environmental Monitoring, Biodiversity Analysis, and Ecosystem Restoration

Acronym: AI4Nature@AVSS

Abstract (for the conference website)

The global climate and biodiversity crises demand a paradigm shift in how we monitor, analyze, and protect our natural world. This workshop, **AI4Nature**, explores the transformative role of advanced visual and signal-based systems in addressing these challenges. In alignment with AVSS 2026's theme of "Expanding Horizons," we move beyond traditional security and surveillance to focus on ecological and environmental applications. The workshop will showcase cutting-edge research in computer vision, deep learning, and multi-modal sensor fusion for the automated monitoring of biodiversity (e.g., species identification, population counting, behavioral analysis), the detection of environmental threats (e.g., pollution, poaching, wildfire, invasive species), and the assessment of ecosystem health. By bringing together computer vision researchers, ecologists, and environmental scientists, AI4Nature aims to foster interdisciplinary collaboration and chart a roadmap for a new generation of intelligent, autonomous systems dedicated to the sustainability and restoration of our planet's ecosystems.

Topics and Relevance

Relevance to AVSS

The AVSS conference has a long-standing tradition of excellence in video and signal-based surveillance. This workshop proposes to expand these core competencies into the rapidly growing and critically important domain of environmental science. The technical challenges are remarkably similar: detecting and tracking objects of interest (animals instead of people) in cluttered scenes, recognizing complex activities (mating, feeding, hunting instead of human interactions), analyzing multi-modal data (audio, thermal, satellite), and deploying systems on edge devices (drones, autonomous robots). AI4Nature directly leverages AVSS's strengths in areas like multi-target tracking, action recognition, anomaly detection, and USV/ROV/UAV-based systems, applying them to solve pressing ecological problems.

Workshop Topics

The workshop will focus on, but is not limited to, the following topics, inspired by the cutting-edge work of the Italian **AI4Nature** project and the broader research community:

1. Computer Vision for Biodiversity Analysis:

- **AI-driven species identification, counting, and tracking** from camera traps, baited remote underwater video systems (Bruvs), drones, and underwater vehicles.
- **Individual animal re-identification** using biometrics (e.g., fur patterns, scars, fin shapes) for population studies.
- **Automated analysis of animal behavior** and social interactions in the wild.

2. Multi-Modal Environmental Monitoring:

- **Sensor fusion** for ecological surveillance, combining video, audio (bioacoustics), thermal, LiDAR, and eDNA data.
- **Anomaly detection** (and early-warning systems) for environmental threats such as poaching, illegal logging, pollution events (e.g., oil spills, microplastics), invasive species, and wildfire ignition.
- **Multimodal foundation models** for understanding complex ecosystem dynamics, linking visual data with climatic and chemical parameters.

3. Autonomous Systems for Ecology:

- **UAV/drone-based monitoring** of remote or inaccessible habitats (forests, coastlines, marine protected areas).
- **Autonomous surface and underwater vehicles** for marine biodiversity assessment and seafloor habitat mapping.
- **Edge AI for real-time, in-situ analysis** on robotic platforms, enabling adaptive mission planning.

4. AI for Ecosystem Health and Restoration:

- **Habitat change detection** using high-resolution satellite and aerial imagery.
- **Assessment of restoration interventions** through automated vegetation and geomorphological analysis.
- **Digital twins of ecosystems** for predictive modeling and scenario analysis to support evidence-based conservation interventions and policy.

This workshop is **highly timely** as the European Union's Green Deal, Biodiversity Strategy 2030, and Nature Restoration Law create an urgent demand for the scalable, accurate, and cost-effective monitoring technologies that our community can provide

Workshop Format (Full Day)

The workshop will be a **full-day (8 hours)** event designed to maximize interaction and community building.

Tentative Schedule:

- **08:45 – 09:00:** Welcome and Opening Remarks (Workshop Organizers)
- **09:00 – 10:30: Session 1: Invited Talks (Vision & Ecology)**
    - *Invited Talk: The Grand challenges of marine biodiversity monitoring", Lisandro Benedetti Cecchi, University of Pisa, Italy*
    - *Invited Talk 2:* CMCC foundation invited talk (TBC)
- **10:30 – 11:00:** Coffee Break
- **11:00 – 12:30: Session 2: Oral Presentations (Peer-Reviewed Papers)**
    - 3-4 high-quality paper presentations selected from submissions, covering novel algorithms and applications.
- **12:30 – 14:00:** Lunch Break
- **14:00 – 15:30: Session 3: Spotlight Talks & Interactive Poster Session**

    - Short "spotlight" presentations (2 minutes each) for all accepted papers, followed by a dedicated poster session to foster in-depth discussions.
- **15:30 – 16:00:** Coffee Break
- **16:00 – 17:30: Session 4: Panel Discussion & Future Roadmap**
    - **Title:** "From Pixels to Policy: How Can Computer Vision Drive the Next Decade of Environmental Action?"
    - **Panelists:** Mix of academic and industry speakers from the day, plus a representative from a major conservation NGO (e.g., WWF) or a European environmental agency (e.g., EEA). The discussion will focus on the pathway from research to real-world impact.
- **17:30 – 18:00: Closing Remarks**

Organizing Committee

Concetto Spampinato, University of Catania, Italy
Marco Milazzo, University of Palermo Italy
Cosimo Distante, CNR, Italy
Paolo Spagnolo, CNR, Italy
Ilyes Benaissa, CNR, Italy

Tentative Program Committee (PC)

TbD

Publicity Plan

We will implement a multi-channel publicity strategy to attract high-quality submissions and attendees from both the computer vision and environmental science communities:

1. **Targeted Mailing Lists:** Distribution of the Call for Papers (CfP) to relevant mailing lists, including CV HUB (CVML, visionscience.com), CVPL, and specialized ecological informatics lists.
2. **AI4Nature Network:** Leveraging the extensive network of the Italian AI4Nature project, which includes over 10 universities and research institutions (CNR, UNIBA, UNISS, UNICT, UNINA, etc.), to promote the workshop.
3. **Social Media:** Promoting the workshop on LinkedIn and X (Twitter) via the organizers' networks and relevant scientific societies (e.g., IEEE SPS, IAPR).
4. **Website:** A dedicated workshop webpage will be created, linked from the main AVSS 2026 site, containing all CfP details, submission instructions, and important dates.

Anticipated Attendance

Given the growing interest in AI for sustainability and the strong community backing from the **AI4Nature project** (which can guarantee a baseline of high-quality submissions and attendees from its 10+ partner institutions), we anticipate:

- **Submissions:** 15-25 full/short papers.
- **Accepted Papers:** 8-12 papers (for oral and poster presentations).
- **Attendees:** 20-40 participants, comprising researchers, students, and practitioners from academia, research centers, and industry. The strong local organizing team in Lecce and the appeal of the topic will help drive attendance.

Special Requirements

- Standard poster boards for the poster session.
- A/V equipment: Projector, screen, microphones, and wireless internet in the main workshop room.